SAMPLING DISTRIBUTION CENTRAL LIMIT THEOREM

Amazon -> 1 M
Can we calculate the average spending of every customer?
Techically --> YES, but practically it will be expensive and time consuming.

"If I select 1000 customers randomly and calculate their average spending?"

Population -->(Random Sample) **Sample ->** (Calculate Statistic) **Sample Mean /Sample Proportion-->** (Repeat sampling many times) **Sampling Distribution**

Population:
The entire group we want to understand.
Example: All customers of an e-commerce company

Sample:
A smaller group selected from that population.
Example: 500 randomly selected customers.

Population Parameter
A numerical value describing the population
μ=Population Mean
p=Population Proportion

Sample Statistic
A numerical value calculated fom the sample.

$\bar{X}$ = Sample Mean
$\hat{p}$ = Sample proportion

One sample gives us one answer. Many samples give us a distibution of answers.

Sample Distribution of The Mean
A sampling distribution is the probability distribution of a statistic calculated from repeated samples of the same size under the same population.

Xbar = (X1+X2+X3+X4+....+Xn)/n

Example:
10 20 30 40 50 Mean 30

Sample size = 2
5C2

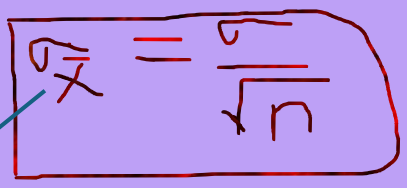
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Possible Samples		Mean
10	20	15
10	30	20
10	40	25
10	50	30
20	30	25
20	40	30
20	50	35
30	40	35
30	50	40
40	50	45
	Mean	30

Mathematical Properties

E(Xbar) = μ

The average of all possible sample means equals to the population mean.



σ = Population Standard Deviation

n = Sample Size

Standard Error

Standard Deviation decribe variablity among individual obsevations.

Standard Error describes variability among sample statistics.

Customer Spending

Average monthly customer spending

Mean (μ) = ₹5,000

SD (σ) = ₹1,200

n = 36

Standard Error = σ /n^1/2 = 1200/36^(1/2)

200

If we repeatedly take random samples of 36 customers, the sample means will typically vary around the mean population of ₹5,000 with a standard error of ₹200

S1	4801	5000-200	5000+200
S2	4900	4800	5200
S3	5100		
S4	5199		

SAMPLING DISTRIBUTION CENTRAL LIMIT THEOREM

Interpretation

Question:

Why don't all sample means equal ₹5,000?

because samples contains different obsevations.

Businss Interpretation:

This is useful for:

1. Customer spending estimation
2. Average Delivery time
3. Average order value.
4. Average salary
5. Average transaction Value.
6. Average Customer satisfaction Score.

Sampling Distribution of Proportions

Population Proportion

$$p = X / N$$

where:

X = number of Successes

N = population Size

Sample Proportion

$$\hat{p} = x/n$$

Example:

Suppose an ecommerce company has:

True conversion Rate = 8%

p = 0.08

Take 200 visitors

suppose 18 purchase

$$\hat{p} = 18/200$$

0.09

So sample conversion rate = 9%

Population is 8% but our sample is 9%

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Why Does Sample Proportion Change?

Taking another 200 visitors

Suppose 14 purchased

$\hat{p} = 14/200$

0.07

Another sample:

$\hat{p} = 0.085$

Another :

$\hat{p} = 0.075$

Therefore:

Different samples produce different sample proportions.

And those proportions form a :

Sampling Distibution Proportion.

Standard Error of Proprtion

$$SE(\hat{p}) = \sqrt{\frac{P(1-P)}{n}}$$

p = population proportion

1- p probability of failure/ non-success

n = Sample size

SE = Expected variability of the sample proportion across repeated samples

Example:

p =0.08

n = 200

$$SE(\hat{p}) = \sqrt{0.08(1-0.08)/200}$$

0.019183

≈0.0192

SE ≈1.92 %

Business Applications

Marketintg

Estimate conversion rate

Team A **A/B Testing**

Team B

$\hat{p}$ A

$\hat{p}$ B

SAMPLING DISTRIBUTION CENTRAL LIMIT THEOREM

Customer Surveys
Percentage of customers satisfied with a product.
Quality Control
Percentage of defective products
Finance
Percentage of customers likely to be default.

Central Limit Theorem

Why need of CLT
Suppose:
Customers purchase amounts are heavily right skewed.
Some customers spend:
₹ 100.00
₹ 200.00
₹ 500.00
₹ 1,000.00
₹ 5,000.00
₹ 50,000.00
So population distribution is not normal

Definition of CLT
When we take sufficiently large random samples from a population and calculate their means,the distribution of those sample means becomes approximately normal.
10000--> 6000
Technically:
For independent and identically distributed observations within finite mean and variance:

$$\bar{X} \approx \left(\mu, \frac{\sigma^2}{n}\right)$$

Or equivalently

$$\bar{X} \approx \left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

depending on whther the second parameter is being expressiod as variance or Standard Deviation.

CLT curve is dependent on sample size

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CLT Mathemtaical Relationship

μ = 1000

σ = 600

n = 100

Then

SE = 600/sqrt(100)

60

Therefore:

Xbar N(1000,60)

The original population can be skewed, but the distribution of sample means is approximately normal when the conditions of CLT are satisfied.

Sample Size demonstration

SE = σ /sqrt(n)

suppose

σ = 20

n = 10

SE = 20/sqrt(10) 6.324555

n = 30

SE = 20/sqrt(30) 3.651484

n = 100

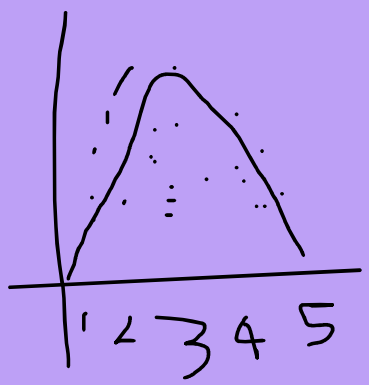
SE =20/sqrt(100) 2

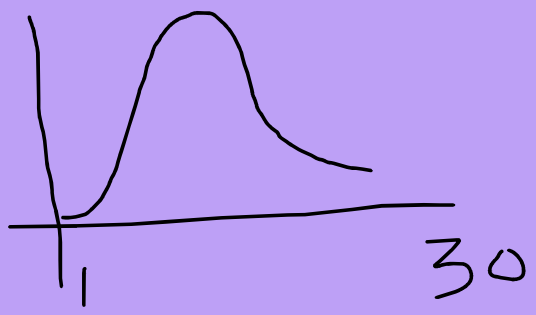
n = 1000 SE = 20/sqrt(1000) 0.632456

Interpretations:

n--> increases then SE --> decreses

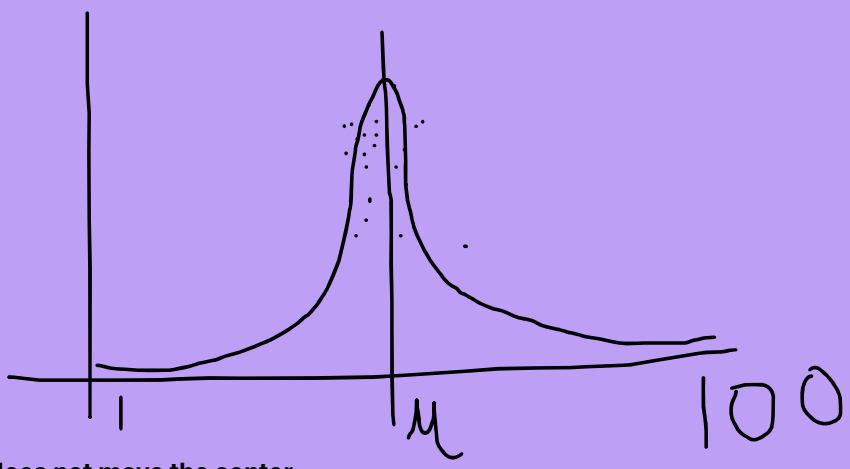
Larger Sample --> more stable sample mean--> narrower sampling distribution.





n = 5 (only 5 values)

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Increasing the sample size does not move the center.

Conditons for CLT

1. Random Sampling

Sample should be selected randomly

Why?

Because biased sampling might produce biaszed conclusion.

2. Indpendence

One observation should not influence other.

Example:

If we survey one customer, their response should not determine the response of the next customer.

3. Sufficiently Large Sample

Common calssrom rule:

n > = 30

For strongly skewed or heavy-tailed populations, a larger sample may be required.

4. Finite Mean and Standard Deviation.

Population Should Have:

$\mu < \infty$

and

$\sigma < \infty$

200

SE 10

190-210